

Parameter	Luna 9 (USSR, 1966)	Mars Pathfinder (NASA, 1997)	MER Spirit / Opportunity (NASA, 2004)
Destination	Moon (Oceanus Procellarum)	Mars (Ares Vallis)	Mars (Gusev / Meridiani)
Total launch mass	1,538 kg	~890 kg	1,062 kg
Lander / shell mass	~1,439 kg (bus+petals)	~360 kg (landed)	365 kg (lander petal structure)
Rover / payload mass	99 kg (ALS capsule)	10.6 kg (Sojourner)	174 kg (Spirit/Opportunity)
Airbag system mass	~15–20 kg (est.)	98 kg (bags + inflators)	~100 kg (bags + inflators)
Airbag type	Non-vented, ejected from bus	Non-vented, 4 bags × 6 lobes	Vented, 4 bags × 6 lobes
Impact velocity	22 km/h (6.1 m/s)	18 m/s (64 km/h)	~14 m/s vertical, ~12 m/s horizontal
Peak decel (g)	~5–10 g (est.)	~40 g (bouncing)	~40 g (bouncing) - (392 m/s ²)
Airbag fabric	Unknown (Soviet-era)	Vectran (Kevlar-class)	Vectran
Bounces	Several (rolled to rest)	15+ (up to 15 m high)	Multiple (rolled to rest)
Lander structure	Bus + 4-petal capsule	Tetrahedral, 4 petals	Tetrahedral, 4 petals (CFRP)
Power	Silver-zinc battery (~100 W)	Solar + battery (180 W peak)	Solar panels (140 W peak)
Comms	Direct to Earth (Jodrell Bank)	UHF direct + relay	UHF via orbiters